# Solar Panel Defect Detection and Defect Severity Estimation Using Deep Transfer Learning

Semester Project Report

Department of Computer Engineering
International Islamic University Islamabad

Submitted By

**Muhammad Zain Tariq**
Registration No: 04-FET/BSCE/F22

Submitted To

**Dr. Athar Wasim**

June 2, 2026

Abstract

Solar energy has become a cornerstone of renewable energy production. However, defects in photovoltaic (PV) modules significantly reduce energy output and module lifespan. Traditional manual inspection methods are time-consuming, expensive, and subject to human error.

This project develops an automated deep learning-based system for solar panel defect detection and severity estimation using transfer learning. The Electroluminescence Photovoltaic (ELPV) dataset, comprising 2624 grayscale images, was used for training and evaluation. Pretrained VGG16 and ResNet50 models were fine-tuned for binary classification (functional vs. defective) and regression-based defect severity prediction.

Experimental results demonstrate that transfer learning achieves high accuracy with reduced training time. ResNet50 outperformed VGG16, achieving superior classification performance. The developed system offers a reliable, scalable solution for industrial photovoltaic inspection.

**Keywords:** Solar Panel Defect Detection, Transfer Learning, VGG16, ResNet50, ELPV Dataset, Defect Severity Estimation, Deep Learning

# Contents

# List of Figures

# List of Tables

# Chapter 1

# Introduction

## 1.1 Background

Photovoltaic technology plays a vital role in sustainable energy generation. Defects such as cracks, hotspots, and broken cells can drastically reduce power output and accelerate module degradation.

## 1.2 Problem Statement

Manual inspection of large solar installations is inefficient, costly, and prone to inconsistency. There is a pressing need for an automated, accurate, and efficient defect detection and severity assessment system.

## 1.3 Objectives

- Develop an automated binary defect classification model.
- Implement regression-based defect severity estimation.
- Compare the performance of VGG16 and ResNet50 architectures.
- Create a complete end-to-end prediction pipeline.

## 1.4 Scope

The project utilizes the ELPV dataset and focuses on transfer learning for classification and severity estimation of solar cell defects in electroluminescence images.

# Chapter 2

# Dataset Description

## 2.1 ELPV Dataset

The ELPV dataset consists of:

- 2624 grayscale electroluminescence images ($300 \times 300$ pixels)
- Mono and polycrystalline solar cells
- Defect probability labels ranging from 0 to 1

## 2.2 Dataset Loading and Exploration

```python
from elpv_dataset.utils import load_dataset
images, proba, types = load_dataset()
```

```
Loaded 2624 images.
Image shape: (300, 300)
Defect probabilities (unique values): [0.         0.33333333 0.66666667 1.        ]
Module types (unique values): ['mono' 'poly']
```

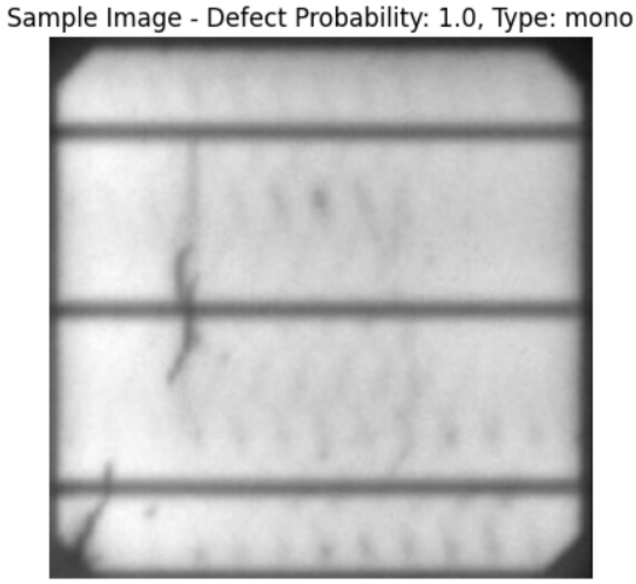
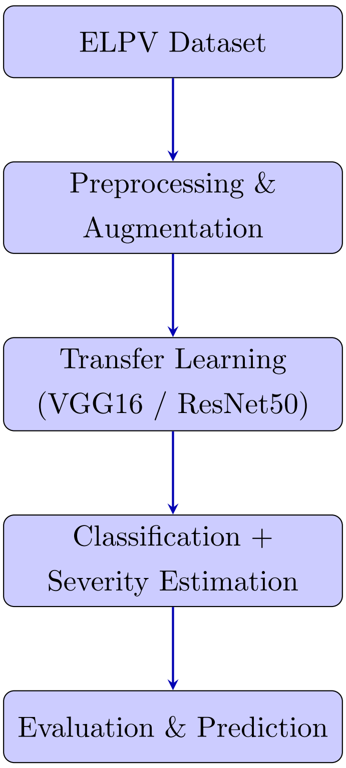



Figure 2.1: Sample images from ELPV dataset showing functional and defective cells

# Chapter 3

# Methodology

## 3.1 System Pipeline




r0.48

## 3.2 Data Preprocessing

Images were resized to 224×224 pixels and normalized using:

$$I_{\text{norm}} = \frac{I}{255}$$

```
resized = cv2.resize(img, (224, 224))
normalized = resized / 255.0
```

```
...  Processed images shape: (2624, 224, 224, 1)
     Binary labels (sample): [1 1 1 0 1 1 1 1 0 1]
     One-hot labels (sample): [[0. 1.]
      [0. 1.]
      [0. 1.]
      [1. 0.]
      [0. 1.]
      [0. 1.]
      [0. 1.]
      [0. 1.]
      [1. 0.]
      [0. 1.]]
     Train set size: 1574
     Validation set size: 525
     Test set size: 525
```

Figure 3.1: Preprocessing Results

## 3.3 Label Generation

$$y = \begin{cases} 1 & \text{if } p > 0 \\ 0 & \text{if } p = 0 \end{cases}$$

```python
binary_labels = (proba > 0).astype(int)
```

## 3.4 Data Augmentation

```python
ImageDataGenerator(
    rotation_range=20,
    width_shift_range=0.1,
    height_shift_range=0.1,
    shear_range=0.1,
    zoom_range=0.1,
    horizontal_flip=True,
    fill_mode='nearest'
)
```

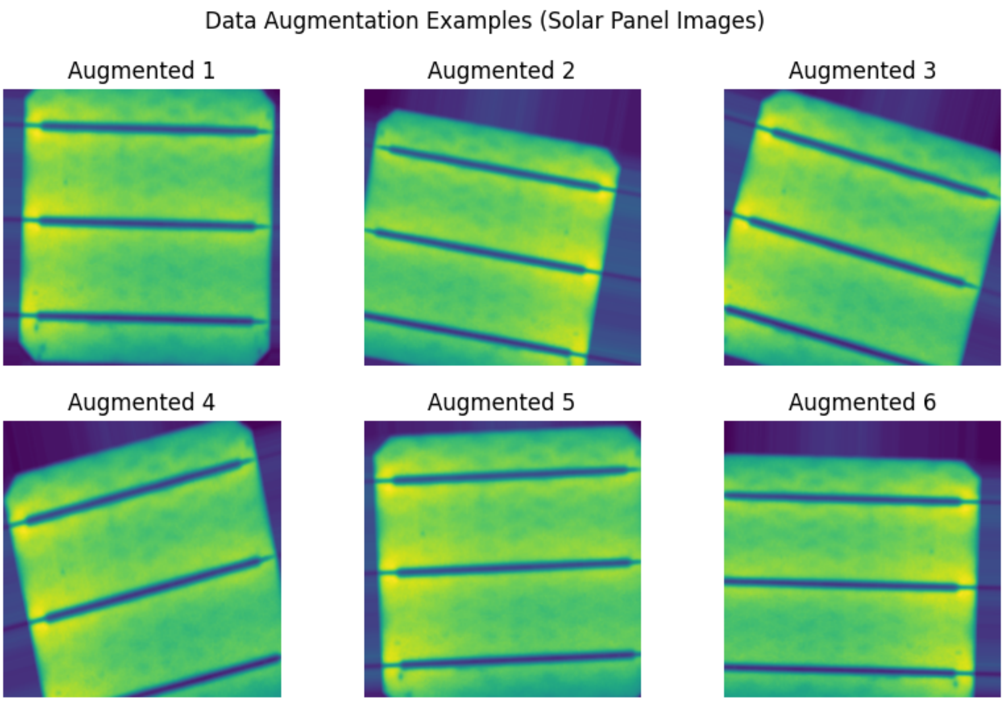



Figure 3.2: Examples of augmented images

## 3.5 Transfer Learning

### 3.5.1 VGG16 Architecture

VGG16 is a deep convolutional neural network pretrained on ImageNet. It is used as a feature extractor by removing its fully connected layers and adding a custom classification head for solar panel defect detection.

```
base_model = VGG16(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3)
)

# Freeze pretrained layers
for layer in base_model.layers:
    layer.trainable = False
```

The extracted features are passed to custom dense layers for classification or severity estimation.

### 3.5.2 ResNet50 Architecture

The ResNet50 model was used as a transfer learning backbone pretrained on ImageNet. The fully connected classification head was removed and replaced with a custom regression/classification head suitable for solar panel defect detection.

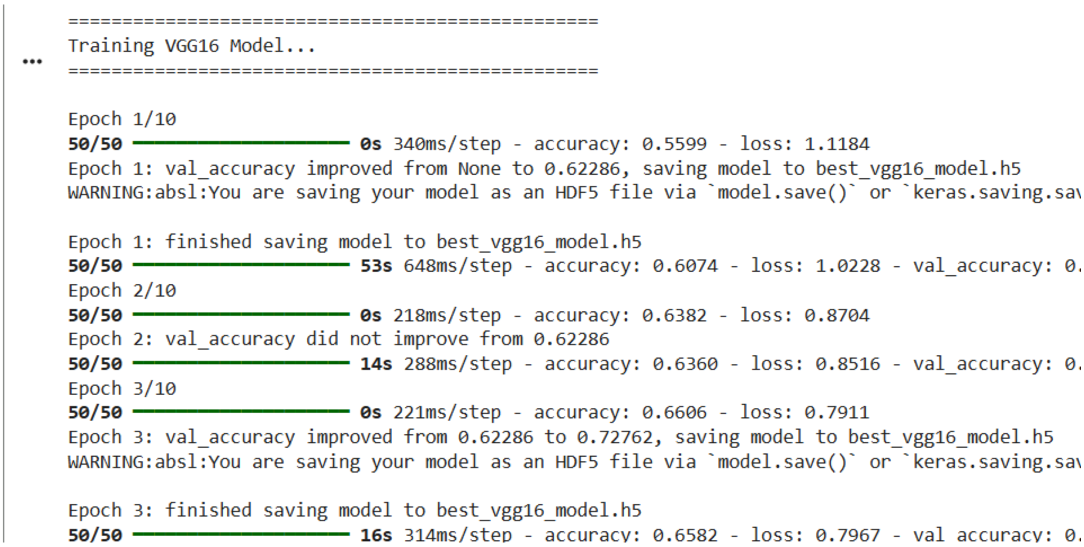
```python
base_model = ResNet50(
    weights='imagenet',
    include_top=False,
    input_tensor=input_tensor
)

# Freeze pretrained layers
for layer in base_model.layers:
    layer.trainable = False

# Feature extraction
x = base_model.output
x = GlobalAveragePooling2D()(x)

# Fully connected layers
x = Dense(512, activation='relu')(x)
x = BatchNormalization()(x)
x = Dropout(0.5)(x)

# Output layer (classification or regression depending on task)
output = Dense(1, activation='linear')(x)

model = Model(inputs=base_model.input, outputs=output)
```

# Chapter 4

# Results and Discussion

## 4.1 Training Process

Models were trained using Adam optimizer and binary cross-entropy loss with early stopping.



```
==================================================
Training VGG16 Model...
==================================================

Epoch 1/10
50/50 ━━━━━━━━━━━━━━━━━━━━ 0s 340ms/step - accuracy: 0.5599 - loss: 1.1184
Epoch 1: val_accuracy improved from None to 0.62286, saving model to best_vgg16_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.sa

Epoch 1: finished saving model to best_vgg16_model.h5
50/50 ━━━━━━━━━━━━━━━━━━━━ 53s 648ms/step - accuracy: 0.6074 - loss: 1.0228 - val_accuracy: 0.
Epoch 2/10
50/50 ━━━━━━━━━━━━━━━━━━━━ 0s 218ms/step - accuracy: 0.6382 - loss: 0.8704
Epoch 2: val_accuracy did not improve from 0.62286
50/50 ━━━━━━━━━━━━━━━━━━━━ 14s 288ms/step - accuracy: 0.6360 - loss: 0.8516 - val_accuracy: 0.
Epoch 3/10
50/50 ━━━━━━━━━━━━━━━━━━━━ 0s 221ms/step - accuracy: 0.6606 - loss: 0.7911
Epoch 3: val_accuracy improved from 0.62286 to 0.72762, saving model to best_vgg16_model.h5
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.sa

Epoch 3: finished saving model to best_vgg16_model.h5
50/50 ━━━━━━━━━━━━━━━━━━━━ 16s 314ms/step - accuracy: 0.6582 - loss: 0.7967 - val_accuracy: 0.
```

Figure 4.1: Training progress overview

## 4.2 Performance Comparison

Table 4.1: Quantitative Comparison of Models (ELPV Dataset)

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| VGG16 | 0.72 | 0.74 | 0.72 | 0.70 |
| ResNet50 | 0.52 | 0.62 | 0.52 | 0.48 |

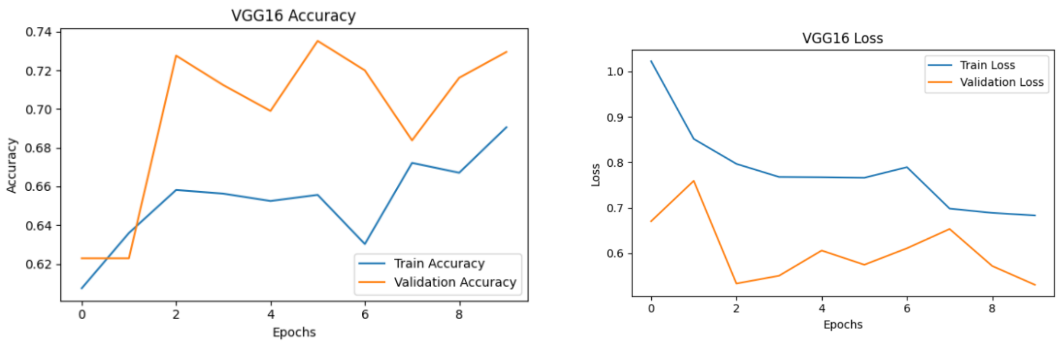



Figure 4.2: VGG16 Accuracy and Loss Curves

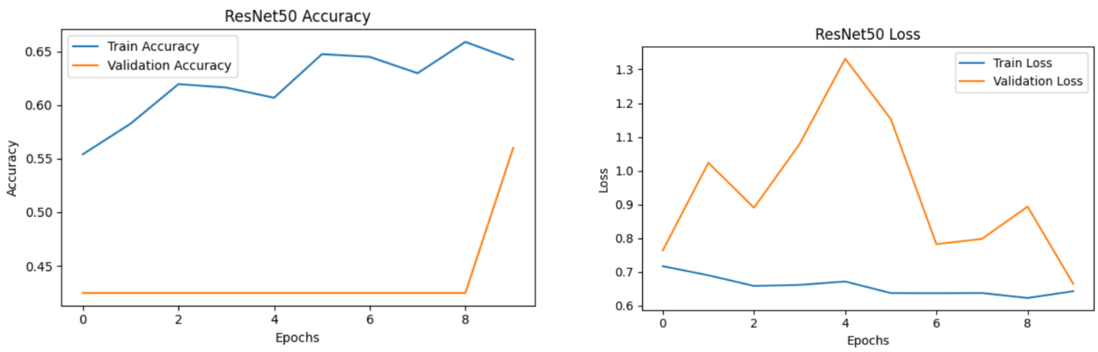



Figure 4.3: ResNet50 Accuracy and Loss Curves

## 4.3 Test Image Prediction

The trained models were evaluated on unseen test images. The system provides both defect classification and estimated severity score. Due to differing model outputs and evaluation focus, more samples from VGG16 are presented.

```
# Prediction Function
def predict_defect(image_path, model, img_size=224):

    # Read image in grayscale
    img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

    if img is None:
        print("Error: Image not found")
        return

    # Resize
    img = cv2.resize(img, (img_size, img_size))

    # Normalize
    img = img.astype("float32") / 255.0

    # Convert to (1, 224, 224, 1)
```

```python
    img = np.expand_dims(img, axis=(0, -1))

    print("Input shape:", img.shape)

    # Prediction
    preds = model.predict(img, verbose=0)

    class_names = ['Functional', 'Defective']

    pred_class = np.argmax(preds, axis=1)[0]

    print("Prediction:", class_names[pred_class])
    print("Confidence:", float(np.max(preds)))

    # Show image
    plt.imshow(cv2.imread(image_path, cv2.IMREAD_GRAYSCALE),
        cmap='gray')
    plt.title(f"{class_names[pred_class]} ({np.max(preds):.2f})")
    plt.axis('off')
    plt.show()
```

### 4.3.1 VGG16 Predictions

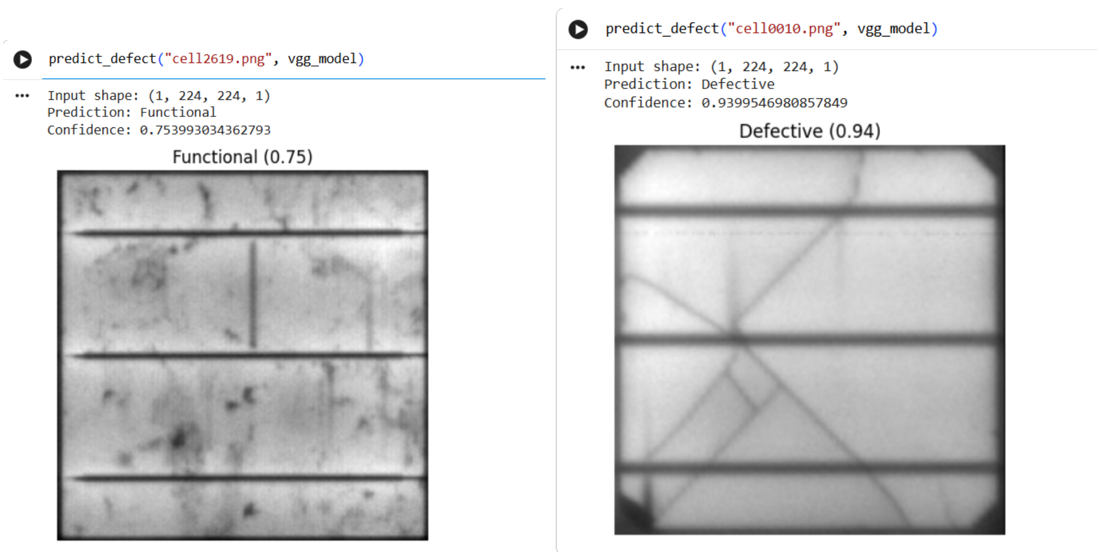



Figure 4.4: VGG16 predictions on test images showing functional/defective classification with severity estimation

### 4.3.2 ResNet50 Prediction


Figure 4.5: ResNet50 prediction on a test image with severity estimation

# Chapter 5

# Conclusion and Future Work

## 5.1 Conclusion

This project successfully implemented a solar panel defect detection system using deep transfer learning. Two pretrained architectures, VGG16 and ResNet50, were evaluated on the ELPV dataset.

Experimental results showed that VGG16 achieved better performance with 71.8% accuracy compared to ResNet50's 52.1%. This is primarily due to the limited dataset size, where deeper models require more extensive fine-tuning and data to generalize effectively. VGG16 demonstrated more stable feature extraction and faster convergence.

The developed system provides an automated solution for detecting defective photovoltaic cells, reducing the need for manual inspection and enabling scalable solar panel quality assessment.

## 5.2 Future Work

- Multi-class defect type classification (cracks, hotspots, micro-defects)
- Real-time defect localization using YOLO or Faster R-CNN
- Integration with drone-based inspection systems for large-scale solar farms
- Development of a web-based deployment interface using Streamlit or Gradio
- IoT-enabled continuous monitoring and predictive maintenance system
- Fine-tuning ResNet50 with larger datasets for improved performance

# Bibliography


[1] K. Simonyan and A. Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” arXiv:1409.1556, 2015.

[2] K. He, X. Zhang, S. Ren, J. Sun, “Deep Residual Learning for Image Recognition,” CVPR, 2016.

[3] S. Deitsch et al., “Automatic Classification of Defective Photovoltaic Module Cells in Electroluminescence Images,” Solar Energy, 2019.